WORKED SOLUTION

STEP I 2013, Question 4

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q4

The key idea

In part (i) each integrand is "a function of something, times the derivative of that something": $\sec^2 x$ is the derivative of $\tan x$, and $\sec x \tan x$ is the derivative of $\sec x$. In part (ii) the extra factor of x or x^2 calls for integration by parts. The trigonometric factor is the one we integrate, and part (i) has just shown us how to do that.

Solution

Part (i)

For the first integral, substitute $u = \tan x$, so $\mathrm{d}u = \sec^2 x \,\mathrm{d}x$. When $x = 0$, $u = 0$, and when $x = \frac{1}{4}\pi$, $u = 1$. Hence

$$\int_0^{\frac{1}{4}\pi} \tan^n x \sec^2 x \,\mathrm{d}x = \int_0^1 u^n \,\mathrm{d}u = \left[\frac{u^{n+1}}{n+1}\right]_0^1 = \frac{1}{n+1},$$

using $n > 0$ so that $0^{n+1} = 0$.

Ask yourself For the second integral, which function's derivative is hiding in $\sec^n x \tan x$?

Since $\dfrac{\mathrm{d}}{\mathrm{d}x} \sec x = \sec x \tan x$, we split off one factor of $\sec x$:

$$\sec^n x \tan x = \sec^{n-1} x \,(\sec x \tan x).$$

Now substitute $u = \sec x$, so $\mathrm{d}u = \sec x \tan x \,\mathrm{d}x$, with $u = 1$ at $x = 0$ and $u = \sqrt{2}$ at $x = \frac{1}{4}\pi$. Since $n > 0$ we may divide by n, and

$$\int_0^{\frac{1}{4}\pi} \sec^n x \tan x \,\mathrm{d}x = \int_1^{\sqrt{2}} u^{n-1} \,\mathrm{d}u = \left[\frac{u^n}{n}\right]_1^{\sqrt{2}} = \frac{(\sqrt{2})^n - 1}{n},$$

as required.

Part (ii)

The first integral. Call it I_1. We integrate by parts, differentiating x and integrating $\sec^4 x \tan x$. The substitution $u = \sec x$ from part (i) gives the antiderivative

$$\int \sec^4 x \tan x \, \mathrm{d}x = \tfrac{1}{4} \sec^4 x + C.$$

(Its definite integral over $[0, \frac{1}{4}\pi]$ is $\frac{1}{4}(4-1) = \frac{3}{4}$, the case $n = 4$ of part (i).) So

$$I_1 = \left[\tfrac{1}{4} x \sec^4 x\right]_0^{\frac{1}{4}\pi} - \frac{1}{4}\int_0^{\frac{1}{4}\pi} \sec^4 x \, \mathrm{d}x = \frac{\pi}{4}\cdot\frac{4}{4} - \frac{1}{4}\int_0^{\frac{1}{4}\pi} \sec^4 x \, \mathrm{d}x,$$

because $\sec^4 \frac{1}{4}\pi = (\sqrt{2})^4 = 4$. For the remaining integral, write $\sec^2 x = 1 + \tan^2 x$:

$$\int_0^{\frac{1}{4}\pi} \sec^4 x \, \mathrm{d}x = \int_0^{\frac{1}{4}\pi} \sec^2 x \, \mathrm{d}x + \int_0^{\frac{1}{4}\pi} \tan^2 x \sec^2 x \, \mathrm{d}x = 1 + \frac{1}{3} = \frac{4}{3},$$

where the first integral is $[\tan x]_0^{\frac{1}{4}\pi} = 1$ and the second is the case $n = 2$ of part (i). Therefore

$$\int_0^{\frac{1}{4}\pi} x \sec^4 x \tan x \, \mathrm{d}x = \frac{\pi}{4} - \frac{1}{3}.$$

The second integral. Call it I_2. Again we differentiate the power of x and integrate $\sec^2 x \tan x$. One antiderivative is $\frac{1}{2}\sec^2 x$, but $\frac{1}{2}\tan^2 x$ works just as well (the two differ by the constant $\frac{1}{2}$), and it is the better choice.

? **Ask yourself** Why prefer $\frac{1}{2}\tan^2 x$ to $\frac{1}{2}\sec^2 x$?

After integrating by parts we will be left with $x \tan^2 x$, and $\tan^2 x = \sec^2 x - 1$ splits into pieces we can integrate. Also, $\tan^2 x$ vanishes at 0, which keeps the boundary terms tidy. (Using $\frac{1}{2}\sec^2 x$ gives the same answer in the end.)

So, with $\tan \frac{1}{4}\pi = 1$,

$$I_2 = \left[\tfrac{1}{2} x^2 \tan^2 x\right]_0^{\frac{1}{4}\pi} - \int_0^{\frac{1}{4}\pi} x \tan^2 x \, \mathrm{d}x = \frac{\pi^2}{32} - \int_0^{\frac{1}{4}\pi} x \tan^2 x \, \mathrm{d}x.$$

Now

$$\int_0^{\frac{1}{4}\pi} x \tan^2 x \, \mathrm{d}x = \int_0^{\frac{1}{4}\pi} x \sec^2 x \, \mathrm{d}x - \int_0^{\frac{1}{4}\pi} x \, \mathrm{d}x = \int_0^{\frac{1}{4}\pi} x \sec^2 x \, \mathrm{d}x - \frac{\pi^2}{32}.$$

Integrating by parts once more, with $\tan x$ as the antiderivative of $\sec^2 x$, and using $\int \tan x \, \mathrm{d}x = \ln \sec x$ (valid since $\sec x > 0$ on this interval):

$$\begin{aligned}\int_0^{\frac{1}{4}\pi} x \sec^2 x \, \mathrm{d}x &= \left[x \tan x\right]_0^{\frac{1}{4}\pi} - \int_0^{\frac{1}{4}\pi} \tan x \, \mathrm{d}x \\ &= \frac{\pi}{4} - \left[\ln \sec x\right]_0^{\frac{1}{4}\pi} = \frac{\pi}{4} - \ln\sqrt{2} = \frac{\pi}{4} - \frac{1}{2}\ln 2.\end{aligned}$$

Putting the pieces together,

$$I_2 = \frac{\pi^2}{32} - \left(\frac{\pi}{4} - \frac{1}{2}\ln 2 - \frac{\pi^2}{32}\right),$$

that is,

$$\int_0^{\frac{1}{4}\pi} x^2 \sec^2 x \tan x \,\mathrm{d}x = \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2.$$

Checks

- Part (i) at $n = 1$: both formulas give $\int_0^{\frac{1}{4}\pi} \tan x \sec^2 x \,\mathrm{d}x = \frac{1}{2}$ and $\int_0^{\frac{1}{4}\pi} \sec x \tan x \,\mathrm{d}x = \sqrt{2} - 1$, which you can confirm directly from $\left[\frac{1}{2}\tan^2 x\right]$ and $[\sec x]$.
- Signs and sizes: both integrands in part (ii) are positive on $(0, \frac{1}{4}\pi)$, and so are the answers, $\frac{\pi}{4} - \frac{1}{3} \approx 0.4521$ and $\frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2}\ln 2 \approx 0.1780$. The second is smaller, as it should be, since there $x^2 \sec^2 x \le x \sec^4 x$ (because $x < 1 < \sec^2 x$).
- A rough bound: $x \sec^4 x \tan x \le \frac{\pi}{4} \sec^4 x \tan x$, so $I_1 \le \frac{\pi}{4} \cdot \frac{3}{4} \approx 0.589$. Our 0.452 is consistent with this. Simpson's rule with a few strips on a calculator gets close to both values.

Common slips

- Substituting $u = \sec x$ in the second integral of part (i) without first splitting off $\sec x \tan x$, which leaves the power as n rather than $n - 1$.
- Forgetting that $\sec \frac{1}{4}\pi = \sqrt{2}$, so that $\sec^4 \frac{1}{4}\pi = 4$ and $\sec^2 \frac{1}{4}\pi = 2$.
- Losing the boundary term $\left[\frac{1}{2}x^2 \tan^2 x\right]$, or getting $\int \tan x \,\mathrm{d}x$ wrong by a sign (it is $\ln \sec x = -\ln \cos x$).

Beyond STEP

The integrals in part (i) are the first steps of the standard theory of integrating $\sec^m x \tan^n x$, where reduction formulae let you lower the powers step by step. Integration by parts with a polynomial factor, as in part (ii), generalises to the "tabular" method and, much later, to the idea of moving derivatives from one function to another that underlies weak solutions of differential equations. For a careful account of why substitution and integration by parts work, see Abbott, *Understanding Analysis*; for many more problems in this style, try Siklos, *Advanced Problems in Mathematics*.

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